ABSTRACT

Child marriage has prevailed as a substantial challenge across the world for ages. It continues to affect the lives of millions of girls each year globally, violating human rights and reducing the opportunities for growth and reinforcing gender inequalities. According to reports of UNICEF 640 million women were married before the age of 18, with one in every 5 girls affected by this issue. Despite considerable advancements towards reducing it, progress remains unequal in different parts of the world, with the highest occurrence in south Asia and south-African countries. Early marriages lead to an extensive number of constraints on young married children, especially girls. It often results in school dropouts, increases health risks, early pregnancy, and limited access to economic and social necessities, retaining the cycle of poverty and dependency, poor maternal and child health outcomes. Eliminating child marriage is the primary goal of the United Nations Sustainable Development Goal (SDG) 5.3, reinforcing the worldwide resolve to terminate this practice by the end of 2030. The bulk of current research on child marriage is geographically limited. Focusing on national or regional datasets such as NFHS or DFS, providing local observations, they fail to capture inter-country differences and worldwide factors of child marriage. Previous studies have largely relied on descriptive, and correlation-based statistical methods, which illustrate trend but lack predictive performance and depth of analysis. Even though numerous global datasets like UNICEF exist, they remain underused for global studies related to child marriage. The usage of machine learning in this domain is very limited, with most researchers focusing on regional investigations. Subsequently, there is no comprehensive research that compares and evaluates the performance of multiple regression and classification models for predicting child marriage risk worldwide. To address this gap, this study aims to develop a data driven robust predictive framework for analyzing the risk of child marriage using machine learning techniques. This research unifies diverse socioeconomic and demographic indicators that are sourced from UNICEF and The World Bank. By merging these datasets, evaluation of multiple regression and classification models were conducted to identify most significant factors affecting child marriage, compare the performance of the models and revealing the best working model to predict child marriage. Multiple regression and classification models were implemented and compared on our dataset, including Linear Regression, Ridge Regression, Lasso Regression, Support Vector Regression (SVR), Decision Tree, Random Forest, Support Vector Classifier (SVC), XGBoost, and CatBoost Classifier. The result revealed that among the regression models, ridge regression performed the best, attaining the lowest Root Mean Square Error (RMSE) and the highest coefficient of R^2 indicating the stability of predicting child marriage. For classification, the CatBoost classifier outdid other models, achieving highest accuracy and ROC-AUC score, differentiating high or low regions prevailing child marriage. This showcases the capability of machine learning in predicting and analyzing global child marriage patterns. This study helps in the development of data-driven policies to accelerate the process of eradication of child marriage globally.

INTRODUCTION

Child Marriage is the union of two individuals with at least one of them under the age of eighteen either formally or informally as per UN/UNICEF (United Nations International Children's Emergency Fund), disproportionately effecting the girl child and mirroring the chronic gender-inequality. "Early" is relative across cultures and the globe. Local traditions refer to "adulthood" and "maturity" far differently. In places such as South Asian and Sub-Saharan, menarche marks the eligibility for marriage often overlooking the chronological age. In certain regions such as Ethiopia, child marriage can be classified as Promissory marriage, referring to a promise or social agreement between two families to marry their children in the future while both the individuals are still minors; marriage at birth or early childhood; Child marriage among girls under the age of 10, and Adolescent marriage of girls between the ages of 10 to 15.

Despite the several attempts of mitigation, the prevalence of child marriage persists. 1 in 5 girls i.e. 750 million women globally have been married before the age of 18. As per UNICEF latest data of the year 2022, 640 million women alive today were married as a child. Nearly 15 million girls under 18 marry every year; South Asia carries the largest burden with India leading in absolute numbers. Everyday 25,000 girls get married. High concentration of recorded cases has been reported in Sub-Saharan Africa (nearly thirty-seven percent), Latin America (approximately twenty-five percent), and South Asia (almost thirty percent). The UN Convention on the Rights of the Child (1989) defines child marriage as a violation. The Universal Declaration of Human Rights (UDHR) mandates that marriage should be based on free and full consent of both the partners. Child marriage inherently violates this provision as most marriages are arranged, with the girl often having never met her husband beforehand. An alternative term like "Child Bride" is often used to romanticize this issue thereby undermining the seriousness.

Child Marriage hinders education often leading to school dropout due to domestic responsibilities and patriarchal ideologies resulting in discontinuation of education. It also affects the autonomy and health resulting in early pregnancy, increasing maternal and infant health risks. Due to biological immaturity, reproductive organs are underdeveloped for child brides to involve in safe sex giving rise to diseases such as sexually transmitted infections (STIs), HIV, cervical cancer. It also increases the risk of complications such as obstetric during pregnancy. Furthermore, it contributes to economic, emotional, financial, psychosocial dependency over their families or partner. Moreover, it limits the thinking abilities and decision-making capacity of the girl child due to subordination. It jeopardizes their self-worth, self-esteem, self-confidence contributing to critical mental health problems building up anxiety, depression and trauma.

Beyond the social sphere, the problem extends to a greater level intensifying a global concern that exerts a significant burden on health and education infrastructures. Early marriage results in early pregnancy and accelerates multiple births due to high fertility rate; however, with limited resources, family struggles to meet the basic needs such as education, healthcare, shelter, food, security. This

adds up to financial burden, reduced human capital, lower workforce participation and rooting poverty.

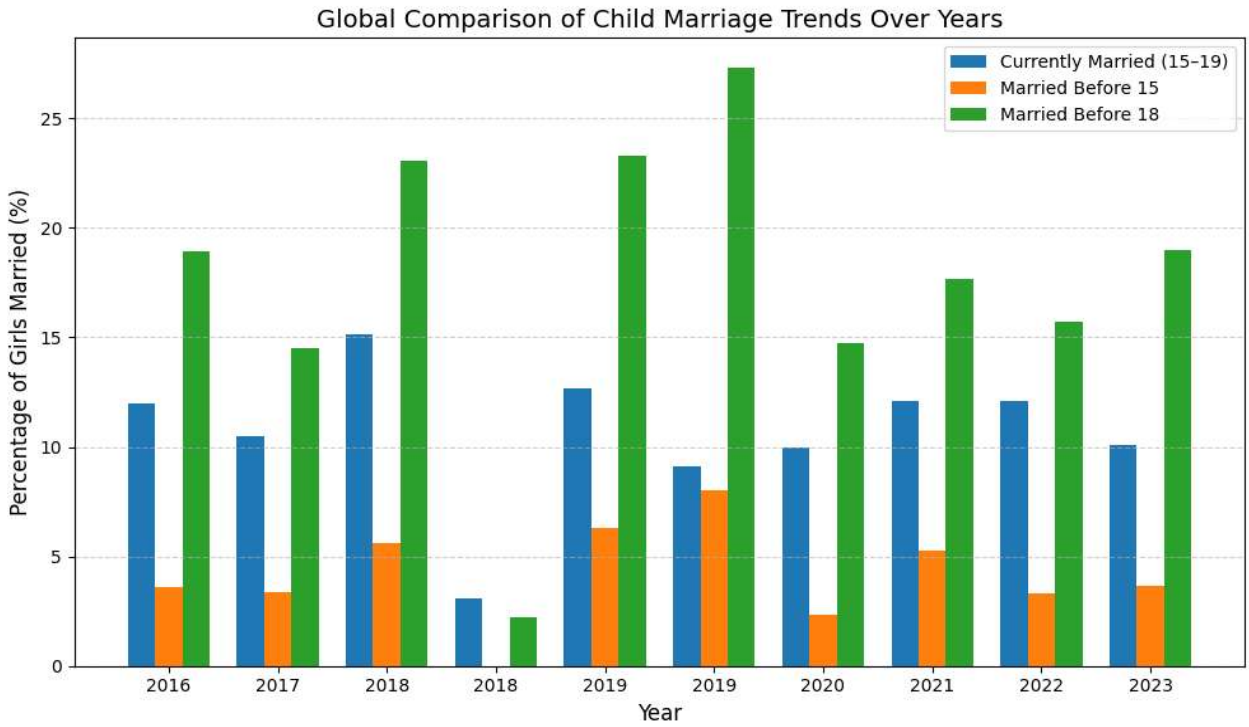

Fig 1.1 Total Child Marriage performed over the years

The above bar graph illustrates the total number of child marriage cases reported over the years from 2016 to 2023 with three indicators- currently married (15-19), married before 15, married before 18. Across each year, the percentage of girls married before 18 are significant larger revealing the seriousness of the issue.

Global commitment under Sustainable Development Goals (SDG 5.3) focuses on eliminating Child Marriage by 2030 and empowering girl child by diminishing gender disparity. This goal also highlights that it is not just a human rights violation, but a development challenge linked with other SDG goals as well such as poverty reduction (SDG 1), quality education (SDG 4), and good health and well-being (SDG 3). Although multiple schemes and legal framework exist such as Prohibition of Child Marriage Act (2006), Protection of Children from Sexual Offences Act i.e. POCSO (2012) in India; along with Ishraq in Egypt, a community-based initiative; Berhane Hewan in Ethiopia, nonetheless their enforcement remains fragile.

Additionally, UNFPA–UNICEF Global Programme to End Child Marriage, launched in 2016 strengthened the community in high-mobilization countries. However, due to limited funding, inadequate monitoring systems, weak legal awareness, the issue remains unresolved.

Upon investigating this concern at depth, it can be concluded that multiple factors and causes are responsible for this outcome. At the individual level, the underlying causes include lack of

education, limited knowledge about pregnancy, legal rights, protection laws and insufficient awareness regarding reproductive health and responsibilities of marriage and motherhood. Girls are often excluded from marriage decisions and often give in out of fear, social pressure or sense of duty. Physical preferences such as men's inclination towards younger girls for beauty, purity, submissiveness are also some of the contributing constituents. Interpersonal factors such as low parental education, parental decision-making dominance, preference for kinship marriage, stigma of surrounding unmarried girls, loss of parents or dysfunctional family are closely associated with child marriage. At community and societal level, determinants include prevailing misconceptions such as marriage shield girls from sexual violence, premarital relations, rumours, unwanted pregnancies, STIs, HIV; it increases their vulnerabilities. Societal norms that place value and family honour on virginity creating immense peer and cultural pressure on the girl. Social rewards act as convincing and desirable option for child marriage. Finally, legislative shortcomings such as weak implementation of marriage age laws, inconsistent marriage laws, gaps in monitoring and enforcement reflect governance failures.

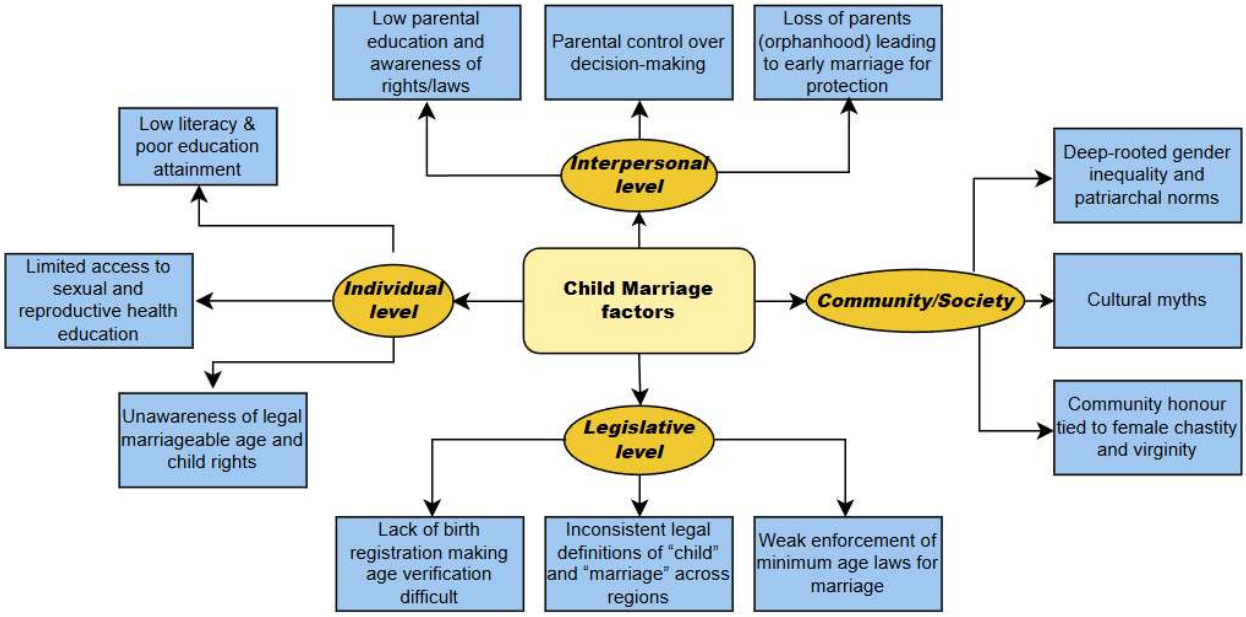

Fig 1.2 Different factors leading to Child Marriage

The above flow chart demonstrates that Child Marriage is driven by varied attributes that can be divided into individual, interpersonal, community/societal and legislative levels. Together, they create social and legal conditions that allow early marriage to perpetuate.

In the recent years, Machine Learning has evolved and emerged significantly as a revolutionary tool for extensive analysis and visualization. Its capability of modelling multifaceted and huge heterogenous data, allows a depth understanding of all the socio-economic and cultural drives such as demographics, education levels, poverty indices, cultural indicators, and geographic attributes

that sustain child marriage. Unlike other traditional statistical methods that centre on understanding relationships between variables and testing hypothesis, machine learning aims at predicting by learning the past patterns. Traditional methods are typically affective for small medium sized datasets, whereas machine learning can analyse large-scale as well as multidimensional datasets.

Traditional Statistical Methods	**Machine Learning Approaches**
Focus on hypothesis testing and parameter estimation	Focus on pattern discovery, feature interactions, and prediction accuracy
Restricted to small or medium-sized datasets	Preferable for large, multidimensional, and heterogeneous data
Requires pre-defined assumptions	Learns from past complex, non-linear, and multidimensional patterns
Models are often theory-driven	Models are data-driven and empirically optimized
Low adaptability to new or unseen data	Highly adaptive and generalizes well to new data
Manual feature selection is required	Can perform automatic feature selection and importance ranking

Table 1.1 Comparative study of traditional statistics vs Machine Learning

Additionally, machine learning facilitates global-level modelling, where data of regional or global areas are combined and used to enhance generalizability and reveal hidden patterns that go beyond local boundaries. It offers a holistic on the determinants or factors that contribute to child marriage, establishing the groundwork for evidence-based interventions and focused policymaking.

This paper, titled "Global Comparison of Regression and Classification Models for Child Marriage Risk Using Machine Learning", aims at comparing different regression and classification models based on their performance to help predict and prevent child marriage at an early stage. By scrutinizing multiple models like Linear Regression, Ridge, Lasso, Support Vector Machines, Decision Trees, XGBoost, and CatBoost, this research intends to determine the best possible algorithm.

Furthermore, it targets to contribute and help policymakers, NGOs, and global organizations by providing a foundation for decision-making based on proper evidence. This study seeks to support and enrich the global efforts towards Sustainable Development Goal (SDG) 5.3 of eliminating child marriage by 2030.

METHODOLOGY

Multiple datasets have been sourced from reliable websites like UNICEF (United Nations Children's Fund) and World Bank because of their worldwide range of socio-economic indicators which aligns with Sustainable Development Goals (SDGs) across the countries. The datasets were obtained in tabular CSV (comma separated values) and Excel formats. The socio-economic variables- *adolescent fertility rate*, *GDP*, *female labor force participation*, *literacy rate*, *poverty rate*, and *rural population,* have been considered based on their impact on and relevance to child marriage. The primary dataset on child marriage prevalence as percentage is extracted from UNICEF addressing women at country-level aged 15-19 whereas the socio-economic variables are collected from the World Bank merged based on country name and reference year. UNICEF global child marriage dataset was used for outcome-related indicator. World Bank has been used for socio-economical indictors that were originally covered from the year 1960 to 2024. Combing both UNICEF and World Bank datasets together provides authoritative and standardization across maximum countries in the globe leading to a well-structured and reliable dataset for global analysis.

To ensure consistency, Country names were standardized across all datasets, and variables were renamed. Irrelevant or duplicate variables were removed such as “Country code”, "Indicator Name”, and “Indicator Code”. Time-based indicators were unified by taking Year_MarriedBefore18 as reference for merging, later renamed as “Year”. The datasets were reshaped from a wide format to a long format.

After preliminary estimation, the dataset was filtered down to eight key variables: CurrentlyMarried15to19, MarriedBefore15, literacy, poverty, rural population, female labor force participation, GDP per capita were appended to the main dataset with MarriedBefore18 as a target variable. MarriedBefore18 has been used as target variable as it aligns with the globally accepted definition of child marriage which refers to, any formal or informal union of two individuals where at least one of them is under the age of 18. It is also consistent with SDG 5.3.1 that aims in eliminating child marriage by the year 2030. The prepared dataset is systematically organized and suitable for further preprocessing.

	A	B	C	D	E	F	G	H	I	J	K
1	Country	CurrentlyMarried15to19	MarriedBefore15	MarriedBefore18	Year	Female_Labor_Force_Participatio	Adolescent_Fertility_Rate	Poverty_Rate	GDP_per_capita	Rural_Population	Literacy_Rate
2	Afghanistan	16.4	9.6	28.7	2023	6.846247344	64.07	47.2	413.76	73.07	22.6
3	Albania	6.7	1.4	11.8	2018	44.28775419	15.46	1	5365.49	39.68	98.3
4	Algeria	3.3	0	3.8	2019	16.95182852	10.33	0.5	4468.45	26.81	74.21
5	Angola	18.2	7.9	30.3	2016	49.74667313	148.9	32.3	1807.95	35.85	80.7

Fig 2.1 First 5 rows of the dataset

The above picture displays the first five rows of the prepared dataset including 11 valid columns- Country, CurrentlyMarried15to19, MarriedBefore15, MarriedBefore18, Year, Female_Labor_Force_Participation, Adolescent_Fertility_Rate, Poverty_Rate, GDP_per_capita, Rural_Population, Literacy_Rate with 116 county-level observations representing different regions across the globe.

Moving on to the data preprocessing, the missing values were identified throughout the datasets including the various socio-economic indicators and main datasets.